Graph-Regularized Optimal Transport for Synchronous Volatility Surface Marking

A Comprehensive Framework with Implementation Guidance

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Abstract

Volatility surface marking for large portfolios requires updating hundreds of surfaces simultaneously in a manner that is internally consistent, respects no-arbitrage constraints, and reflects the sparse stream of live market quotes. We develop a framework that achieves this by working in the space of risk-neutral distributions rather than pointwise implied volatilities.

The approach rests on three pillars: (i) the one-dimensional L^2-Wasserstein geometry, which makes quantile functions the natural coordinate system for probability measures; (ii) the Log-Quantile-Density (LQD) parametrisation, which guarantees butterfly no-arbitrage by construction; (iii) a directed-graph regulariser that propagates observed shocks to unobserved nodes as a discrete harmonic extension.

We provide the full mathematical development, concrete basis specifications, semi-analytic Jacobian formulae, calendar arbitrage algorithms, and graph construction heuristics. We then identify and address four structural tensions between the mathematical framework and market reality: the re-normalisation feedback loop that couples shape and scale updates, the basis boundary paradox that pits tail fidelity against wing noise, the structural causality problem when idiosyncratic events dominate the graph signal, and the misfit introduced by post-hoc calendar arbitrage repair. For each, we develop concrete solutions suitable for production implementation within a real-time latency budget.

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Part I: The Mathematical Framework

1 Motivation and Context

1.1 The marking problem

A bank's equity derivatives desk may warehouse exposure on 500+ underlyings, each requiring a volatility surface across 8–12 maturities and 20–40 strikes. At any given moment, only a small fraction of these points is directly observable from the market (liquid index options, a handful of single-stock tenors). The rest must be *inferred*—and they must be updated *synchronously* whenever the liquid part of the market moves.

Trader's Intuition

Think of it this way: when SPX 3-month 25-delta put vol moves 2 points, you need your entire book re-marked within seconds—not just the SPX surface, but every correlated single-stock surface, every maturity, every strike. The question is: what is the principled way to propagate that shock?

Traditional approaches—parallel-shifting the smile, applying a "sticky-delta" or "sticky-strike" rule, or running independent SABR/SVI fits—have well-known shortcomings:

- They treat each surface in isolation, ignoring cross-asset consistency.
- They operate on pointwise implied volatilities, which are one-dimensional projections of the full risk-neutral distribution. Moving a single vol point can break butterfly or calendar arbitrage elsewhere on the surface.
- They lack a coherent framework for deciding *how much* an unobserved surface should move when a correlated liquid surface moves.

1.2 The key idea: propagate distributions, not vol points

This framework takes a different route. Instead of working with implied volatilities $\sigma(K,T)$, we work with the full risk-neutral measure $\mu_{i,T}$ for each asset–maturity node. The quantile function $Q_{i,T}(u)$ provides a linearisation of the Wasserstein metric, the LQD map $\psi_{i,T}(u) = \log Q'_{i,T}(u)$ ensures monotonicity, and a directed graph encodes which nodes influence which. The result is a single convex optimisation that simultaneously:

1. fits observed quotes,
2. propagates information to unobserved nodes,
3. preserves no-arbitrage (butterfly by construction; calendar via constrained optimisation or projection).

Trader's Intuition

Instead of thinking "SPX vol moved, so I shift all single-stock smiles by some fraction," the framework says: "The SPX risk-neutral density reshaped in a specific way—here is what that implies for every other density in the universe, given their structural relationships."

2 The State Space: Log-Forward Returns

Definition 2.1 (Log-forward return)**.** For asset i at maturity T, let $S_{i,T}$ be the terminal price and $F_{i,T}$ the forward. Under the risk-neutral measure $\mathbb{Q}$, define:

$$X_{i,T} \;=\; \log\left(\frac{S_{i,T}}{F_{i,T}}\right). \tag{1}$$

The law of $X_{i,T}$ under $\mathbb{Q}$ is denoted $\mu_{i,T}$.

The forward-normalisation removes two nuisances at once: the risk-free drift and the absolute price level. What remains is the "pure optionality" content of the measure. Since $\mathbb{E}^{\mathbb{Q}}[S_{i,T}] = F_{i,T}$ by definition of the forward, the martingale constraint reads:

$$\mathbb{E}^{\mathbb{Q}}\left[e^{X_{i,T}}\right] = 1. \tag{2}$$

This is not a vacuous normalisation: it fixes the "centre of mass" of e^X under the risk-neutral measure. When we perturb the shape of the distribution later, we must always re-impose (2)—otherwise we introduce a forward bias, which shows up as a spurious drift in delta hedging.

Definition 2.2 (Quantile function)**.** Let $\Phi_{i,T}$ be the CDF of $\mu_{i,T}$. The quantile function is:

$$Q_{i,T}(u) = \inf\{x \in \mathbb{R} : \Phi_{i,T}(x) \geq u\}, \quad u \in (0,1). \tag{3}$$

The quantile function is a complete characterisation of the measure—it encodes the same information as the CDF, the density, or the characteristic function, but it has a crucial geometric advantage that we exploit in Section 5.

Assumption 2.3 (Regularity)**.** Throughout this paper, we assume that every $\mu_{i,T}$ has:

(a) a density $f_{i,T} \in L^1(\mathbb{R}) \cap L^\infty(\mathbb{R})$ with respect to Lebesgue measure,

(b) finite fourth moment: $\int_{\mathbb{R}} x^4 \, d\mu_{i,T}(x) < \infty$.

Condition (a) ensures that $Q_{i,T}$ is strictly increasing and continuous, so that $Q'_{i,T}(u) > 0$ a.e. and the LQD (Section 4) is well defined. Condition (b) ensures the Wasserstein tangent-space approximation is well controlled and that moments of the reconstructed distribution remain finite. For equity options, both conditions are comfortably satisfied by any standard model (Black-Scholes, Heston, SABR, SVI, etc.).

3 Coordinate Separation: Location, Scale, Shape

3.1 The decomposition

Definition 3.1 (Location–Scale–Shape)**.** Decompose the quantile function as:

$$Q_{i,T}(u) \;=\; m_{i,T} + s_{i,T}\,\widetilde{Q}_{i,T}(u), \tag{4}$$

where:

$$m_{i,T} = \int_0^1 Q_{i,T}(u)\,du \qquad \text{(location: mean of } X_{i,T}\text{)}, \tag{5}$$

$$s_{i,T} = \left(\int_0^1 \left(Q_{i,T}(u) - m_{i,T}\right)^2 du\right)^{1/2} \qquad \text{(scale: std. dev. of } X_{i,T}\text{)}, \tag{6}$$

$$\widetilde{Q}_{i,T}(u) = \frac{Q_{i,T}(u) - m_{i,T}}{s_{i,T}} \qquad \text{(shape: unit-variance quantile)}. \tag{7}$$

The shape satisfies the normalisations:

$$\int_0^1 \widetilde{Q}_{i,T}(u)\,du = 0, \qquad \int_0^1 \widetilde{Q}_{i,T}(u)^2\,du = 1. \tag{8}$$

3.2 Why separate?

The separation serves both mathematical and practical purposes:

1. **Prevents mode confusion.** Without separation, a uniform 2-vol-point parallel shift of the smile would be absorbed into the "shape" coordinates, distorting what should be a pure scale move. By isolating $s_{i,T}$ (overall vol level) from $\widetilde{Q}_{i,T}$ (skew, kurtosis, higher-order shape), each component can be modelled and propagated on its own terms.
2. **Different dynamics.** Scale (s) tends to be driven by broad market factors (VIX level, sector beta); shape ($\widetilde{Q}$) encodes idiosyncratic smile features (earnings skew, dividend discreteness, tail risk). Separating them allows the graph to carry different weights for scale propagation vs. shape propagation.
3. **Numerical conditioning.** If all nodes operate on unit-variance shapes, the basis expansion coefficients live on comparable scales, improving the condition number of the linear system.

Note that the decomposition (4) defines a quotient by the affine group $(m, s) \mapsto m + s \cdot (\,)$. The shape manifold $\{\widetilde{Q} : \int \widetilde{Q} = 0,\ \int \widetilde{Q}^2 = 1\}$ is the unit sphere in a centred L^2 subspace. A finite perturbation of the LQD changes s as well as $\widetilde{Q}$: the shape and scale are coupled through the reconstruction. The consequences of this coupling—and the feedback loop it creates—are analysed in detail in Section 13.

4 The Log-Quantile-Density (LQD)

4.1 Definition and reconstruction

Definition 4.1 (LQD)**.** For a quantile function Q with a.e. positive derivative, the Log-Quantile-Density is:

$$\psi(u) = \log Q'(u), \qquad u \in (0, 1). \tag{9}$$

The reconstruction (inverse map) is:

$$Q(u) = a + \int_0^u e^{\psi(s)}\,ds, \tag{10}$$

where a is a constant fixed by the location constraint or the martingale condition.

The LQD is the centrepiece of the framework. Its fundamental property is that *any* function $\psi \in L^1(0, 1)$, when exponentiated and integrated, produces a strictly increasing quantile function. There is no additional constraint to impose.

Proposition 4.2 (Butterfly no-arbitrage by construction)**.** *If $\psi \in L^1(0, 1)$, then the quantile Q defined by (10) satisfies $Q'(u) = e^{\psi(u)} > 0$ a.e., so Q is strictly increasing. Consequently, the risk-neutral density is non-negative: the surface is free of butterfly arbitrage.*

Proof. $Q'(u) = e^{\psi(u)} > 0$ for a.e. $u \in (0, 1)$ since exp is everywhere positive. The density of the corresponding measure in strike space is $f(K) = \partial^2 C/\partial K^2 \geq 0$, which is equivalent to the quantile being non-decreasing. Strict positivity of Q' gives strict monotonicity, hence a strictly positive density wherever it is defined. □

Trader's Intuition

This is the main payoff of the LQD construction: you can freely adjust the function $\psi(u)$ without ever worrying about butterfly arbitrage. Compare this to working directly with implied volatilities or call prices, where every perturbation must be checked for density positivity. Here, positivity is *automatic*.

4.2 Boundary behaviour and basis specification

For realistic distributions (log-normal tails, Student-t tails, power-law tails), the quantile derivative $Q'(u)$ diverges as $u \to 0^+$ or $u \to 1^-$, so $\psi(u) \to +\infty$ at both boundaries. This means $\psi \notin L^\infty(0,1)$, and any finite basis expansion $\psi \approx \sum_m \beta_m \phi_m$ must use basis functions that can represent this boundary growth. Furthermore, near the boundaries e^ψ is large, so the linearisation $e^{\psi_0+\delta\psi} \approx e^{\psi_0}(1+\delta\psi)$ has larger residuals unless $\delta\psi$ is small relative to 1.

We use a **log-augmented orthogonal polynomial basis**. Let $\{L_k\}_{k\geq 0}$ be the (shifted) Legendre polynomials on $(0,1)$. Define:

$$\phi_0(u) = 1, \qquad \phi_1(u) = -\log u, \qquad \phi_2(u) = -\log(1-u), \qquad \phi_m(u) = L_{m-2}(u), \;\; m \geq 3. \tag{11}$$

The rationale for each component:

- ϕ_0: constant level (absorbed by the integration constant a).
- ϕ_1: captures the left-tail singularity. For a distribution with left-tail exponent α, we have $\psi(u) \sim \alpha \cdot (-\log u)$ as $u \to 0^+$.
- ϕ_2: captures the right-tail singularity, analogous role for the right tail.
- ϕ_m, $m \geq 3$: smooth interior deformations (skew, kurtosis, higher cumulants).

The coefficients β_1, β_2 of the tail basis functions must satisfy $\beta_1 > 0$ and $\beta_2 > 0$ to produce integrable tails (i.e. for e^ψ to remain integrable near the boundaries). The treatment of these tail coefficients—in particular the tension between wing fidelity and noise amplification—is a subtle problem addressed in Section 14.

Trader's Intuition

The tail basis functions ϕ_1, ϕ_2 directly correspond to what traders call "tail richness." A large β_1 means the left tail of the risk-neutral density is heavy (puts are expensive); a large β_2 means the right tail is heavy (calls are expensive). The Legendre terms capture interior shape: ϕ_3 is roughly skew, ϕ_4 is roughly excess kurtosis, and so on.

4.3 From LQD back to implied volatility

The chain from ψ to $\sigma_{\mathrm{IV}}(K,T)$ is:

$$\psi(u) \xrightarrow{\exp + \int} Q(u) \xrightarrow{u=\Phi(K)} f(K) = \frac{1}{Q'(\Phi(K))} \xrightarrow{\text{Breeden–Litzenberger}} C(K) \xrightarrow{\mathrm{BS}^{-1}} \sigma_{\mathrm{IV}}(K).$$

Explicitly:

$$C(K,T) = e^{-rT} \int_0^1 \big(F_{i,T}\, e^{Q_{i,T}(u)} - K\big)^+ du, \tag{12}$$

$$\sigma_{\mathrm{IV}}(K,T) = \mathrm{BS}^{-1}\big(C(K,T),\, F_{i,T},\, K,\, T,\, r\big). \tag{13}$$

5 Wasserstein Geometry

5.1 The W_2 distance in quantile coordinates

Definition 5.1 (L^2-Wasserstein distance)**.** For $\mu, \nu \in \mathcal{P}_2(\mathbb{R})$ (probability measures with finite second moment):

$$W_2^2(\mu, \nu) = \int_0^1 \big(Q_\mu(u) - Q_\nu(u)\big)^2 \, du. \tag{14}$$

This is the defining identity of one-dimensional optimal transport. In higher dimensions, computing W_2 requires solving a linear programme; in 1D, it is just an L^2 norm of quantile differences.

Trader's Intuition

W_2 measures how much "probability mass needs to move" to transform one distribution into another. Two distributions can have the same mean and variance but very different W_2 distance (because their shapes differ). By working with W_2, we capture the full distributional change, not just moment changes.

5.2 Linearity and the shock representation

With the location–scale–shape decomposition (4):

$$W_2^2(\mu^{\text{prior}}, \mu^{\text{new}}) = (\Delta m)^2 + (\Delta s)^2 + s_{\text{prior}}^2 \, \|\Delta \widetilde{Q}\|_{L^2}^2 + \text{cross terms}, \tag{15}$$

where the cross terms vanish when either $\Delta m = 0$ or $\Delta s = 0$. In practice, we handle m and s updates separately from shape updates, making the geometry essentially Pythagorean.

A "shock" to the shape is:

$$\Delta \widetilde{Q}_{i,T}(u) = \widetilde{Q}_{i,T}^{\text{new}}(u) - \widetilde{Q}_{i,T}^{\text{prior}}(u). \tag{16}$$

In quantile space, this is a vector in $L^2(0,1)$—the Wasserstein tangent space. The graph regulariser will penalise $\|\Delta \widetilde{Q}_v - \sum_w W_{vw} \Delta \widetilde{Q}_w\|^2$: the distributional change at node v should be close to the weighted average of changes at its neighbours.

5.3 First-order approximation in LQD coordinates

Proposition 5.2 (Wasserstein metric in coefficient space)**.** *Fix location and scale. Let $\psi^{(0)}$ be the prior LQD and $\psi = \psi^{(0)} + \sum_m \beta_m \phi_m$. Then:*

$$W_2^2(\mu^{\text{prior}}, \mu^{\text{new}}) \;\approx\; s^2 \, \boldsymbol{\beta}^\top G \, \boldsymbol{\beta}, \tag{17}$$

where the Gram matrix is:

$$G_{jk} = \int_0^1 \left(\int_0^u e^{\psi^{(0)}(t)} \phi_j(t) \, dt \right) \left(\int_0^u e^{\psi^{(0)}(t)} \phi_k(t) \, dt \right) du. \tag{18}$$

Proof. Linearise the reconstruction: $\widetilde{Q}^{\text{new}}(u) - \widetilde{Q}^{\text{prior}}(u) \approx \int_0^u e^{\psi^{(0)}(t)} \sum_m \beta_m \phi_m(t) \, dt$. Substituting into $\|\Delta \widetilde{Q}\|_{L^2}^2$ and exchanging the order of integration yields (17)–(18). □

The Gram matrix G depends on the prior $\psi^{(0)}$, so the induced metric on coefficient space is node-dependent and prior-dependent. The Euclidean penalty $\|\boldsymbol{\beta}\|^2$ used in the inverse problem (Section 7) does not exactly coincide with the W_2 metric. As a practical compromise, we use a diagonal rescaling: replace $\mathbf{I}_M$ in the smoothness penalty with $\text{diag}(g_{11}, \ldots, g_{MM})$ where $g_{mm} = \int_0^1 (\int_0^u e^{\bar{\psi}(t)} \phi_m(t) \, dt)^2 \, du$ for an "average" prior $\bar{\psi}$ across nodes. This is cheap and approximately Wasserstein-correct.

6 The Directed Graph

6.1 Setup and properties

Definition 6.1 (Volatility influence graph)**.** Let $\mathcal{V} = \{(i, T)\}$ be the set of all asset–maturity nodes. The influence structure is encoded by a matrix $\mathbf{W} \in \mathbb{R}^{|\mathcal{V}|\times|\mathcal{V}|}$ satisfying:

(W1) $W_{vv} = 0$ (no self-influence).

(W2) $W_{vw} \geq 0$ (non-negative influence).

(W3) $\sum_w W_{vw} \leq 1$ (sub-stochastic: total influence ≤ 1).

The *residual* $\alpha_v = 1 - \sum_w W_{vw} \in [0, 1]$ represents the node's "self-trust"—how much it anchors to its own prior.

Trader's Intuition

The interpretation is simple: $W_{vw} = 0.3$ means that when we have no direct observation for node v, it should absorb 30% of whatever happened at node w. The self-trust α_v is how much it sticks to its prior. For a liquid node where we have direct quotes, α_v is effectively overridden by the data fidelity term.

Non-negativity **(W2)** is essential: without it, the graph Laplacian $\mathbf{L} = \mathbf{I} - \mathbf{W}$ is not an M-matrix, the discrete maximum principle fails, and the harmonic interpolation (Section 8) can produce oscillatory or non-physical solutions. Modelling anti-correlation between asset classes requires a different penalty structure (see Section 17).

The graph Laplacian is:

$$\mathbf{L} = \mathbf{I} - \mathbf{W}. \tag{19}$$

Lemma 6.2 (Properties of $\mathbf{L}$)**.** *Under **(W1)–(W3)**:*

(i) $\mathbf{L}$ *is a weakly diagonally dominant M-matrix with positive diagonal and non-positive off-diagonal entries.*

(ii) All eigenvalues λ *of* $\mathbf{L}$ *satisfy* $\mathrm{Re}(\lambda) \geq 0$.

(iii) If there exists at least one node with $\alpha_v > 0$ *(strictly sub-stochastic row) and the graph is weakly connected, then* $\mathbf{L}$ *is non-singular.*

Proof. (i)–(ii): immediate from Gershgorin's theorem. (iii): Strict diagonal dominance at one node, combined with irreducibility (weak connectivity), propagates to all nodes by Taussky's theorem, giving non-singularity. □

If *all* row sums equal 1 (purely stochastic $\mathbf{W}$), then $\mathbf{L}\mathbf{1} = \mathbf{0}$: the Laplacian has a non-trivial kernel and the inverse problem has no unique solution without additional regularisation. The Tikhonov term $\eta \|\boldsymbol{\beta}\|^2$ in the objective (Section 7) resolves this, but one should be aware that in the stochastic limit the solution increasingly depends on the Tikhonov parameter rather than the graph structure. We recommend always setting $\alpha_v \geq 0.05$ at every node.

6.2 Graph construction heuristics

The specification of $\mathbf{W}$ is a modelling choice. Here are concrete guidelines.

1. Same-asset maturity ladder. For asset i, maturities $T_1 < \cdots < T_N$:

$$W_{(i,T_n),(i,T_{n\pm1})} \propto \frac{1}{|T_{n\pm1} - T_n|}. \tag{20}$$

Neighbouring maturities influence each other inversely proportional to their distance. This captures term-structure smoothness.

2. Cross-asset at same maturity. For assets $i \neq j$ at the same maturity T:

$$W_{(i,T),(j,T)} \propto |\rho_{ij}| \cdot \sqrt{\frac{\ell_j}{\ell_i + \ell_j}}, \tag{21}$$

where ρ_{ij} is the historical (or implied) correlation and ℓ_j is a liquidity score (e.g. average daily option volume). The liquidity weighting ensures that liquid names influence illiquid ones more than the reverse.

3. Index–constituent link. If i is a constituent of index I with weight w_i^I:

$$W_{(i,T),(I,T)} \propto w_i^I \cdot \rho_{iI}. \tag{22}$$

This is typically the strongest link in the graph: the index drives its constituents.

4. Row normalisation. After setting all raw weights, normalise each row:

$$W_{vw} \leftarrow (1 - \alpha_v) \frac{W_{vw}}{\sum_{w'} W_{vw'}},$$

so that $\sum_w W_{vw} = 1 - \alpha_v$. The self-trust α_v can be set per node: $\alpha_v = 0.1$ for liquid nodes (quotes will dominate anyway), $\alpha_v = 0.3$ for illiquid nodes (more anchoring to prior). The dynamic adjustment of α_v around idiosyncratic events is treated in Section 15.

Practical Remark

The graph does not need to be large or dense. In practice, each node typically has 5–15 neighbours (2 maturity neighbours + 3–10 cross-asset peers + 1–2 index links). The resulting $\mathbf{W}$ is very sparse, which makes the linear system in Section 7 cheap to solve.

7 The Inverse Problem

7.1 Observation model

Partition the nodes into *observed* ($\mathcal{V}_{\text{obs}}$) and *unobserved* ($\mathcal{V}_{\text{unobs}}$). For each observed node $v \in \mathcal{V}_{\text{obs}}$, we have n_v option quotes (implied volatilities at strikes $K_1, \ldots, K_{n_v}$). The "shock" vector is:

$$y_v = \left(\sigma_{\text{IV}}^{\text{mkt}}(K_k, T) - \sigma_{\text{IV}}^{\text{prior}}(K_k, T)\right)_{k=1}^{n_v} \in \mathbb{R}^{n_v}. \tag{23}$$

7.2 LQD basis expansion

The perturbation to the LQD at node v is:

$$\Delta\psi_v(u) = \sum_{m=1}^{M} \beta_{v,m}\, \phi_m(u), \tag{24}$$

where $\{\phi_m\}$ is the basis (11). Stack all coefficients: $\boldsymbol{\beta} \in \mathbb{R}^{|\mathcal{V}|\cdot M}$.

7.3 The Jacobian

The map from LQD coefficients to option quotes is:

$$\beta_v \xrightarrow{\text{LQD update}} \psi_v^{(0)} + \Delta\psi_v \xrightarrow{\exp+\int} Q_v \xrightarrow{(12)} C_v(K_k) \xrightarrow{\text{BS}^{-1}} \sigma_{\text{IV},v}(K_k).$$

The Jacobian at node v is $A_v \in \mathbb{R}^{n_v \times M}$:

$$(A_v)_{k,m} = \left.\frac{\partial\, \sigma_{\text{IV},v}(K_k)}{\partial \beta_{v,m}}\right|_{\boldsymbol{\beta}=\mathbf{0}}. \tag{25}$$

By the chain rule, differentiating the call price integral (12) under the integral sign (justified by dominated convergence under Assumption 2.3):

$$\frac{\partial C_v(K)}{\partial \beta_{v,m}} = e^{-rT}\, F_v\, s_v \int_{u^*(K)}^{1} e^{m_v + s_v \widetilde{Q}_v(u)} \underbrace{\left(\int_0^u e^{\psi_v^{(0)}(t)} \phi_m(t)\, dt\right)}_{\partial \widetilde{Q}_v / \partial \beta_{v,m}} du, \tag{26}$$

where $u^*(K) = \Phi_v^{-1}(K)$ is the quantile level corresponding to strike K. The vol Jacobian follows via the Black-Scholes vega:

$$(A_v)_{k,m} = \frac{1}{\mathcal{V}_v(K_k)} \frac{\partial C_v(K_k)}{\partial \beta_{v,m}}, \tag{27}$$

where $\mathcal{V}_v(K_k)$ is the Black-Scholes vega at that strike. This semi-analytic formula avoids finite differences and is numerically stable even near the boundaries of $(0, 1)$.

7.4 The variational objective

Definition 7.1 (Objective function)**.**

$$\boxed{J(\boldsymbol{\beta}) = \underbrace{\sum_{v \in \mathcal{V}_{\text{obs}}} \|y_v - A_v \beta_v\|^2_{\Sigma_v^{-1}}}_{\text{data fidelity}} + \lambda \underbrace{\sum_v \left\| \beta_v - \sum_w W_{vw} \beta_w \right\|^2}_{\text{graph consistency}} + \eta \underbrace{\sum_v \beta_v^\top \Omega\, \beta_v}_{\text{smoothness / regularisation}}} \tag{28}$$

The three terms are:

1. **Data fidelity:** fit the observed quotes. The weighting Σ_v^{-1} is the inverse noise covariance. In practice, $\Sigma_v = \text{diag}(\sigma_1^2, \ldots, \sigma_{n_v}^2)$ with $\sigma_k = \frac{1}{2}(\text{ask}_k - \text{bid}_k)$, preventing wide-spread wing quotes from dominating the fit.
2. **Graph consistency:** the LQD perturbation at each node should be close to the influence-weighted average of its neighbours. This is the propagation mechanism.
3. **Smoothness:** the matrix Ω penalises roughness in the LQD perturbation. A natural choice is $\Omega = \text{diag}(\omega_1, \ldots, \omega_M)$ with small weights on the first few basis functions (let the data speak) and large weights on high-order terms (regularise oscillations). The specific treatment of the tail coefficients ω_1, ω_2 is discussed in Section 14.

7.5 Matrix form and normal equations

In block notation, the graph consistency term is:

$$\sum_v \left\| \beta_v - \sum_w W_{vw} \beta_w \right\|^2 = \|(\mathbf{L} \otimes \mathbf{I}_M)\, \boldsymbol{\beta}\|^2.$$

Note that for a directed graph, we write the penalty as a sum over row-wise residuals (not $\boldsymbol{\beta}^\top(\mathbf{L}^\top\mathbf{L} \otimes \mathbf{I}_M)\boldsymbol{\beta}$), which ensures the exact harmonic property at all unobserved nodes (see Section 8). The minimiser satisfies:

$$\boxed{\left[\mathbf{A}^\top\boldsymbol{\Sigma}^{-1}\mathbf{A} + \lambda\,\mathbf{H}_{\text{graph}} + \eta\,(\mathbf{I}_{|\mathcal{V}|} \otimes \Omega)\right]\hat{\boldsymbol{\beta}} = \mathbf{A}^\top\boldsymbol{\Sigma}^{-1}\mathbf{y},} \tag{29}$$

where $\mathbf{A} = \text{diag}(A_{v_1}, \dots, A_{v_n})$ is block-diagonal (only observed nodes have non-empty blocks), $\boldsymbol{\Sigma} = \text{diag}(\Sigma_{v_1}, \dots)$, $\mathbf{y}$ is the stacked shock vector, and $\mathbf{H}_{\text{graph}}$ is the Hessian of the row-wise penalty. For symmetric graphs, $\mathbf{H}_{\text{graph}} = \mathbf{L}^\top\mathbf{L} \otimes \mathbf{I}_M$; for directed graphs, it takes the form derived from the per-node residuals (see Section 8 for the distinction).

Practical Remark

The system (29) has dimension $|\mathcal{V}| \cdot M$. For 200 assets $\times$ 10 maturities $\times$ 5 basis functions = 10,000 unknowns. The matrix is sparse (the graph is sparse), symmetric positive definite (given $\eta > 0$ and $\Omega \succ 0$), and has a natural block structure. Preconditioned conjugate gradient with a block-diagonal preconditioner converges in $\sim$20 iterations, taking $<$ 50ms on modern hardware.

7.6 Hyperparameter interpretation

Parameter	Role	Typical range
λ	Graph coupling strength	10^{-2} to 10^{1}
η	Smoothness / Tikhonov damping	10^{-4} to 10^{-1}
α_v	Self-trust (in $\mathbf{W}$)	0.05 to 0.30
Σ_v	Quote uncertainty	half bid–ask spread

Trader's Intuition

Large λ means "trust the graph": unobserved nodes move a lot when their neighbours move. Small λ means "stick to your prior." The right balance is found by backtesting: simulate removing one liquid node from the observed set, propagate from neighbours, and check the prediction error. This is standard leave-one-out cross-validation.

8 The Harmonic Property

The graph consistency term has a clean interpretation for unobserved nodes.

Proposition 8.1 (Harmonic interpolation). *If node v has no observations (A_v absent) and $\eta = 0$, the optimality condition for β_v reduces to:*

$$\hat{\beta}_v = \sum_w W_{vw}\,\hat{\beta}_w. \tag{30}$$

With Tikhonov damping $\eta > 0$:

$$\hat{\beta}_v = \frac{\lambda}{\lambda + \eta\,\omega_{\text{eff}}} \sum_w W_{vw}\,\hat{\beta}_w. \tag{31}$$

The unobserved node's state is a shrinkage of its neighbours' weighted average toward zero (the prior).

Equation (30) holds exactly when the penalty is written as a sum of per-node (row-wise) residuals $\sum_v \|\beta_v - \sum_w W_{vw}\beta_w\|^2$. If one instead uses the matrix-form penalty $\boldsymbol{\beta}^\top(\mathbf{L}^\top\mathbf{L}\otimes\mathbf{I}_M)\boldsymbol{\beta}$, the first-order condition for v also involves the rows of $\mathbf{L}$ with non-zero entries in the v-column: nodes that v influences (its "children"). For "hub" nodes (e.g. a liquid index that influences many stocks), the exact first-order condition becomes:

$$\left(1 + \sum_{w:W_{wv}>0} W_{wv}^2\right)\hat{\beta}_v = \sum_w W_{vw}\hat{\beta}_w + \sum_{w:W_{wv}>0} W_{wv}\left(\hat{\beta}_w - \sum_{u\neq v} W_{wu}\hat{\beta}_u\right). \tag{32}$$

The additional terms pull $\hat{\beta}_v$ toward consistency with its children, not just its parents. For a liquid index node influencing many single stocks, this effect is quantitatively significant. We therefore adopt the row-wise penalty throughout, which gives the clean harmonic equation (30) at all unobserved nodes. A practical reinforcement: make the graph nearly acyclic (index $\to$ constituents, not the reverse) so that the index is a leaf in the influence sense.

9 Single-Maturity Simplification

When we fix a single maturity T and work only with the cross-sectional graph, the problem simplifies to a classical harmonic extension.

Partition the nodes into observed (O) and unobserved (U). Write the influence matrix in block form: $\mathbf{W}_{UU}$, $\mathbf{W}_{UO}$. For unobserved nodes with no data fidelity term, the harmonic condition gives:

$$\boxed{\hat{\boldsymbol{\beta}}_U = \underbrace{(\mathbf{I} - \mathbf{W}_{UU})^{-1}}_{=\mathbf{L}_{UU}^{-1}}\,\mathbf{W}_{UO}\,\hat{\boldsymbol{\beta}}_O.} \tag{33}$$

Trader's Intuition

This is the punchline for the single-maturity case: solve for the observed nodes first (they fit their own quotes subject to smoothness), then propagate to unobserved nodes by a single matrix multiply. The matrix $(\mathbf{I} - \mathbf{W}_{UU})^{-1}\mathbf{W}_{UO}$ is the "propagation matrix"—it tells you exactly how many vol points each unobserved stock moves per vol point of each observed stock. You can precompute it and apply it in microseconds.

If $\mathbf{W}_{UU}$ is strictly sub-stochastic ($\alpha_v > 0$ for all unobserved nodes), then $\rho(\mathbf{W}_{UU}) < 1$ and the Neumann series converges:

$$(\mathbf{I} - \mathbf{W}_{UU})^{-1} = \sum_{k=0}^{\infty} \mathbf{W}_{UU}^k.$$

The terms have an intuitive interpretation: $k = 0$ is direct influence from observed nodes; $k = 1$ is one hop through unobserved intermediaries; $k = 2$ is two hops. The series decays geometrically and 3–5 terms suffice in practice.

9.1 Worked example: 4 assets

Example 9.1 (Index + 3 stocks)**.** Consider an index I and three stocks A, B, C at a single maturity, with a two-dimensional LQD basis ($M = 2$: one skew mode, one kurtosis mode).

Influence matrix (after normalisation with $\alpha = 0.1$):

$$\mathbf{W} = \begin{pmatrix} 0 & 0.36 & 0.27 & 0.18 \\ 0.45 & 0 & 0.18 & 0.09 \\ 0.54 & 0.18 & 0 & 0.09 \\ 0.36 & 0.27 & 0.18 & 0 \end{pmatrix}$$

Row sums are all $0.81 < 1$, confirming sub-stochasticity.

Suppose the index and stock A are observed with shocks:

$$\hat{\beta}_I = \begin{pmatrix} 0.10 \\ 0.02 \end{pmatrix}, \qquad \hat{\beta}_A = \begin{pmatrix} 0.05 \\ -0.01 \end{pmatrix}.$$

For the unobserved nodes B, C:

$$\mathbf{W}_{UU} = \begin{pmatrix} 0 & 0.09 \\ 0.18 & 0 \end{pmatrix}, \quad \mathbf{W}_{UO} = \begin{pmatrix} 0.54 & 0.18 \\ 0.36 & 0.27 \end{pmatrix}.$$

$$(\mathbf{I} - \mathbf{W}_{UU})^{-1} \approx \begin{pmatrix} 1.016 & 0.091 \\ 0.183 & 1.016 \end{pmatrix}.$$

Computing $\mathbf{W}_{UO}\hat{\boldsymbol{\beta}}_O$:

$$(\mathbf{W}_{UO}\hat{\boldsymbol{\beta}}_O)_B = 0.54 \begin{pmatrix} 0.10 \\ 0.02 \end{pmatrix} + 0.18 \begin{pmatrix} 0.05 \\ -0.01 \end{pmatrix} = \begin{pmatrix} 0.063 \\ 0.009 \end{pmatrix},$$
$$(\mathbf{W}_{UO}\hat{\boldsymbol{\beta}}_O)_C = 0.36 \begin{pmatrix} 0.10 \\ 0.02 \end{pmatrix} + 0.27 \begin{pmatrix} 0.05 \\ -0.01 \end{pmatrix} = \begin{pmatrix} 0.050 \\ 0.004 \end{pmatrix}.$$

After the $(\mathbf{I} - \mathbf{W}_{UU})^{-1}$ correction:

$$\hat{\beta}_B \approx \begin{pmatrix} 0.068 \\ 0.020 \end{pmatrix}, \qquad \hat{\beta}_C \approx \begin{pmatrix} 0.062 \\ 0.006 \end{pmatrix}.$$

Interpretation: Stock B (high index correlation, $W_{BI} = 0.54$) picks up most of the index skew shock. Stock C picks up a moderate amount, blending index and stock A influences. The second component (kurtosis) is small for both, since the index and A had opposing signs ($+0.02$ vs -0.01) and the influence averages them.

10 Reconstruction and Arbitrage Enforcement

10.1 LQD to quantile

For each node v, the updated LQD is:

$$\psi_v^{\text{new}}(u) = \psi_v^{(0)}(u) + \sum_m \hat{\beta}_{v,m}\, \phi_m(u). \tag{34}$$

The updated shape quantile is:

$$\widetilde{Q}_v^{\text{new}}(u) = C_v + \int_0^u e^{\psi_v^{\text{new}}(s)}\, ds, \tag{35}$$

with C_v chosen to enforce $\int_0^1 \widetilde{Q}_v^{\text{new}} = 0$.

10.2 Re-normalisation

After reconstructing $\widetilde{Q}_v^{\text{new}}$, its L^2-norm will generally no longer be exactly 1 (the LQD perturbation changes the variance of the shape). We re-normalise:

$$s_v^{\text{new}} = s_v^{(0)} \cdot \|\widetilde{Q}_v^{\text{new}}\|_{L^2}, \tag{36}$$

$$\widetilde{Q}_v^{\text{new}} \leftarrow \frac{\widetilde{Q}_v^{\text{new}}}{\|\widetilde{Q}_v^{\text{new}}\|_{L^2}}. \tag{37}$$

This ensures that the shape–scale decomposition remains consistent. The interaction between this re-normalisation and the Jacobian computed at the prior is a source of coupling that we analyse in Section 13.

10.3 Martingale condition

The location m_v^{new} is determined by the constraint (2):

$$m_v^{\text{new}} = -\log \int_0^1 e^{s_v^{\text{new}}\, \widetilde{Q}_v^{\text{new}}(u)}\, du. \tag{38}$$

Proposition 10.1 (Martingale-consistent LQD propagation)**.** *The location parameter m_v is a non-linear functional of the basis coefficients:*

$$m_v(\boldsymbol{\beta}) = -\log \int_0^1 \exp\Big(s_v(\boldsymbol{\beta}) \cdot \widetilde{Q}_v(\boldsymbol{\beta}, u)\Big)\, du, \tag{39}$$

where both s_v and $\widetilde{Q}_v$ depend on $\boldsymbol{\beta}$ through the reconstruction and re-normalisation chain. This non-linearity means that the Jacobian A_v (computed at $\boldsymbol{\beta} = 0$) does not account for the location shift induced by a shape change: the solver optimises the fit at the prior's location, but the reconstructed surface sits at a different location.

This is the primary non-linearity in the system and the main justification for the outer Gauss–Newton iteration in the algorithm (Section 11).

10.4 Butterfly arbitrage: guaranteed

By Proposition 4.2, the reconstructed quantile is strictly increasing for *any* choice of $\hat{\boldsymbol{\beta}}$. Butterfly no-arbitrage is automatic and never needs to be checked or enforced.

10.5 Calendar arbitrage

Calendar no-arbitrage requires that for a single asset with maturities $T_1 < \cdots < T_N$:

$$\mu_{T_1} \leq_{\text{cx}} \mu_{T_2} \leq_{\text{cx}} \cdots \leq_{\text{cx}} \mu_{T_N}, \tag{40}$$

where $\leq_{\text{cx}}$ denotes the convex order. In terms of call prices: $C(K, T_n) \leq C(K, T_{n+1})$ for all K.

In quantile space, $\mu \leq_{\text{cx}} \nu$ is equivalent to:

$$\int_0^p Q_\mu(u)\, du \geq \int_0^p Q_\nu(u)\, du \quad \forall\, p \in (0,1), \qquad \int_0^1 Q_\mu = \int_0^1 Q_\nu. \tag{41}$$

A basic approach is post-hoc projection: for each asset and consecutive maturity pair, compute the integrated quantile functions on a grid, detect violations, and apply a pool-adjacent-violators

(isotonic regression) algorithm to the cumulative quantile differences. This runs in $O(J)$ per maturity pair.

However, this post-hoc approach has a structural limitation: it operates outside the optimisation, and large projections can destroy the fit to market quotes. We address this in Section 16 with a constrained QP formulation that enforces calendar monotonicity *within* the solve.

11 The Full Algorithm

We assemble all components into a complete algorithm.

Algorithm: The Synchronous Marker

Inputs:

- Prior surfaces $\{\psi_v^{(0)}, m_v^{(0)}, s_v^{(0)}\}$ for all $v \in \mathcal{V}$.
- Influence matrix $\mathbf{W}$ (precomputed; Section 6.2).
- Live market quotes for observed nodes $\mathcal{V}_{\text{obs}}$.
- Event schedule $\mathcal{E}$ for dynamic self-trust (Section 15).

Step 0: Dynamic Self-Trust Update. For nodes with scheduled idiosyncratic events, increase α_v per Section 15. Re-normalise affected rows of $\mathbf{W}$.
Step 1: Shock Computation. For each $v \in \mathcal{V}_{\text{obs}}$, compute $y_v = \sigma_{\text{IV}}^{\text{mkt}} - \sigma_{\text{IV}}^{\text{prior}}$.
Step 2: Jacobian Assembly. For each observed node, compute A_v via (26). Assemble block-diagonal $\mathbf{A}$.
Step 3: Constrained Solve. Solve the constrained QP (Section 16):

$$\min_{\boldsymbol{\beta}} J(\boldsymbol{\beta}) \quad \text{s.t.} \quad \beta_{v,1} \geq \epsilon, \;\; \beta_{v,2} \geq \epsilon, \;\; \text{calendar constraints.}$$

If calendar constraints are handled post-hoc (simpler implementation), solve the unconstrained system (29) by preconditioned CG.
Step 4: LQD Update. For each $v \in \mathcal{V}$: $\psi_v^{\text{new}} = \psi_v^{(0)} + \sum_m \hat{\beta}_{v,m}\phi_m$.
Step 5: Quantile Reconstruction + Re-normalisation. Compute $\widetilde{Q}_v^{\text{new}}$ via (35). Enforce (8) via (36)–(37).
Step 6: Martingale Enforcement. Fix m_v^{new} via (38).
Step 7: Calendar Projection (if not in Step 3). For each asset, enforce convex-order monotonicity.
Step 8: Convergence Check. If residual $\max_v \|y_v - A_v\hat{\beta}_v\|_\infty / \|y_v\|_\infty > \epsilon$: update prior $\psi_v^{(0)} \leftarrow \psi_v^{\text{new}}$, recompute Jacobians, return to Step 2.
Step 9: Output. Convert Q_v^{new} to implied volatilities via (12)–(13).

Practical Remark

Performance budget (200 assets $\times$ 10 maturities, $M = 5$, 200-point quantile grid):

Step	**Time**
Self-trust update + row renormalisation	< 1 ms
Jacobian assembly (all observed nodes)	5–10 ms
CG solve or QP (2000 nodes $\times$ 5 coeff.)	10–50 ms
Quantile reconstruction + re-normalisation	2–5 ms
Martingale fix + calendar projection	1–3 ms
IV output (all strikes, all nodes)	5–10 ms
Total (single iteration)	**25–80 ms**
Total (3 iterations)	**75–240 ms**

This is within the latency budget for real-time marking ($<$ 500ms). The QP solve is more expensive than unconstrained CG but remains feasible with a sparse solver such as OSQP.

12 Formal Properties

Theorem 12.1 (Well-posedness). *Suppose $\eta > 0$ and $\Omega \succ 0$. Then the objective (28) has a unique minimiser $\hat{\boldsymbol{\beta}}$.*

Proof. The objective is a strictly convex quadratic (the Hessian is the sum of positive semi-definite terms, with $\eta \mathbf{I} \otimes \Omega$ providing strict positivity). □

Theorem 12.2 (Butterfly no-arbitrage). *For any $\hat{\boldsymbol{\beta}} \in \mathbb{R}^{|\mathcal{V}| \cdot M}$, the reconstructed risk-neutral density at every node is non-negative.*

Proof. Proposition 4.2. □

Theorem 12.3 (Information propagation). *In the single-maturity, purely graph-regularised case ($\eta = 0$), the solution at unobserved nodes satisfies $\hat{\boldsymbol{\beta}}_U = (\mathbf{I} - \mathbf{W}_{UU})^{-1} \mathbf{W}_{UO} \hat{\boldsymbol{\beta}}_O$. The propagation matrix $P = (\mathbf{I} - \mathbf{W}_{UU})^{-1} \mathbf{W}_{UO}$ satisfies $P \geq 0$ (entrywise) and $P\mathbf{1} \leq \mathbf{1}$: the propagation is a contraction.*

Proof. $(\mathbf{I} - \mathbf{W}_{UU})^{-1} = \sum_{k \geq 0} \mathbf{W}_{UU}^k \geq 0$ since $\mathbf{W}_{UU} \geq 0$. The bound $P\mathbf{1} \leq \mathbf{1}$ follows from sub-stochasticity. □

Part II: Where Geometry Meets Market Reality

The framework developed in Part I is mathematically elegant: a convex quadratic objective, a guaranteed butterfly-free reconstruction, and a closed-form harmonic propagation. We now examine four structural tensions where the mathematical machinery collides with market reality, and develop concrete solutions for each.

13 The Re-Normalisation Feedback Loop

13.1 The problem

The Jacobian A_v (Section 7.3) is computed at the prior $\boldsymbol{\beta} = \mathbf{0}$, which implicitly assumes a specific scale $s_v = s_v^{(0)}$. The solver finds $\hat{\boldsymbol{\beta}}$ that minimises the linearised data fidelity $\|y_v - A_v\hat{\beta}_v\|^2$. But the subsequent reconstruction chain—exponentiate, integrate, re-normalise (Section 10.2), fix the martingale (Section 10.3)—operates non-linearly, and the re-normalisation step changes s_v. If the graph propagation suggests a large shape change (e.g. a stress event widens the distribution), the variance of $\widetilde{Q}_v^{\text{new}}$ may deviate significantly from 1, making the post-solve re-normalisation a substantial correction.

The result: the surface that emerges from reconstruction may be measurably different from the one the solver thought it was optimising. The outer Gauss–Newton loop (Step 8 in Section 11) is designed to handle this, but for large shocks the iteration may be slow or, in pathological cases, oscillate.

Implementation Warning

Concretely, suppose a stress event widens the SPX distribution by 20%. The solver, working at the prior scale, finds $\hat{\beta}$ that fits the observed quotes. After reconstruction and re-normalisation, s_v jumps by a factor proportional to the variance change, shifting all option prices. On the next iteration, the new Jacobian is computed at the updated scale, producing a correction in the opposite direction. If the scale–shape coupling is strong, this ping-pong may take 5–7 iterations to settle.

13.2 Solution: Manifold-Constrained QP

The root cause is that the solver operates in the tangent space while the constraints (unit variance, martingale) live on a manifold. A cleaner approach is to linearise the constraints and include them in the QP.

Scale constraint. The variance of the perturbed shape (to first order in $\boldsymbol{\beta}$) is:

$$\|\widetilde{Q}_v^{(0)} + \delta\widetilde{Q}_v\|_{L^2}^2 \approx 1 + 2\int_0^1 \widetilde{Q}_v^{(0)}(u)\,\delta\widetilde{Q}_v(u)\,du. \tag{42}$$

For the shape to remain unit-variance to first order, we need:

$$\int_0^1 \widetilde{Q}_v^{(0)}(u)\,\delta\widetilde{Q}_v(u)\,du = 0 \qquad \Longleftrightarrow \qquad \mathbf{c}_v^\top \beta_v = 0, \tag{43}$$

where $\mathbf{c}_v \in \mathbb{R}^M$ with $(c_v)_m = \int_0^1 \widetilde{Q}_v^{(0)}(u) \int_0^u e^{\psi_v^{(0)}(t)}\phi_m(t)\,dt\,du$.

Martingale constraint. Linearising $\int_0^1 e^{Q_v(u)} du = 1$ around the prior:

$$\int_0^1 e^{Q_v^{(0)}(u)}\, s_v\, \delta\widetilde{Q}_v(u)\, du = 0 \qquad \Longleftrightarrow \qquad \mathbf{d}_v^\top \beta_v = 0, \tag{44}$$

where $\mathbf{d}_v$ has a similar integral form weighted by $e^{Q_v^{(0)}(u)}$.

Including these as equality constraints in the QP transforms the problem from:

$$\min_{\boldsymbol{\beta}}\ J(\boldsymbol{\beta}) \qquad \text{to} \qquad \min_{\boldsymbol{\beta}}\ J(\boldsymbol{\beta}) \quad \text{s.t.} \quad \mathbf{c}_v^\top \beta_v = 0,\ \ \mathbf{d}_v^\top \beta_v = 0\ \ \forall v.$$

This ensures that the "data fidelity" term is minimised on the manifold of martingale-compliant, unit-variance distributions. The re-normalisation and martingale-fix steps become small second-order corrections rather than first-order jumps, and the outer iteration converges in 1–2 steps even for large shocks.

Practical Remark

Adding $2|\mathcal{V}|$ equality constraints to a 10,000-variable QP is trivial for any modern sparse QP solver. The constraint vectors $\mathbf{c}_v, \mathbf{d}_v$ are computed from the same integrals already needed for the Jacobian, so the marginal cost is negligible.

14 The Basis Boundary Paradox

14.1 The tension

The log-augmented basis functions $\phi_1(u) = -\log u$ and $\phi_2(u) = -\log(1-u)$ capture the tail singularities of $\psi(u)$. But the tails are precisely where two competing forces collide:

1. **Wing fidelity.** Traders care deeply about implied volatility asymptotics at deep out-of-the-money strikes. The Roger Lee moment formula gives $\sigma^2_{\mathrm{IV}}(K)\,T \to 2(\tilde{p}-1)$ for large $\log(K/F)$, where $\tilde{p}$ is the critical moment exponent of the risk-neutral measure. If the market exhibits fat tails beyond what $-\log u$ can represent, the basis will under-fit the wings.
2. **Wing noise.** Deep OTM quotes are wide (bid–ask spreads of 2–5 vol points) and often stale. If the tail coefficients β_1, β_2 are too free, they absorb noise from these illiquid quotes, producing non-physical "hooks" in the implied volatility surface.

14.2 Tail-specific regularisation

We resolve this by introducing an **informative prior on the tail coefficients**. In the smoothness matrix Ω (the regulariser in (28)), set:

$$\Omega = \mathrm{diag}(\omega_1, \omega_2, \omega_3, \ldots, \omega_M) \quad \text{with} \quad \omega_1 = \omega_2 = \omega_{\mathrm{tail}} \gg \omega_m \text{ for } m \geq 3. \tag{45}$$

The effect: the tail coefficients are "pulled" toward the prior's tail decay rate. Only strong, consistent market evidence (a genuine richening of the wings across multiple strikes) can overcome the pull. Noisy individual OTM quotes are filtered out.

Trader's Intuition

Think of ω_{tail} as an expression of "how much do I trust the prior's tails versus the market's wings." Setting it high means "I believe my prior's tail shape is right unless the market emphatically disagrees." Setting it low means "let the market move the tails freely." In

calm markets, $\omega_{\text{tail}} = 1$ (moderate anchoring) works well; in tail events (COVID, SVB), reduce it to 0.1 so the tails can adjust.

14.3 Higher-order tail basis functions

For markets with genuinely extreme tail behaviour (e.g. implied vol surfaces with 60+ vol on 10-delta puts), the $-\log u$ singularity may be insufficient. One can augment the basis with:

$$\phi_{\mathrm{L},\alpha}(u) = (-\log u)^{\alpha}, \qquad \phi_{\mathrm{R},\alpha}(u) = (-\log(1-u))^{\alpha}, \tag{46}$$

for $\alpha > 1$. These produce heavier-tailed distributions. The exponent α can be calibrated from the prior surface's tail index or from the Lee moment formula applied to the observed wing quotes.

Implementation Warning

Adding higher-order tail basis functions increases the dimensionality of β_v and requires correspondingly stronger regularisation to avoid overfitting. We recommend at most $M = 7$ total basis functions (ϕ_0, ϕ_1, ϕ_2 plus 4 Legendre terms) for production use; adding tail augmentation should be reserved for nodes with genuinely unusual tail behaviour.

15 Structural Causality vs. Observation Lag

15.1 The problem

The graph propagation is inherently *reactive*: it pushes information from observed nodes to unobserved ones. This works well when the dominant risk factor is market-wide (a VIX spike rippling through the universe). It breaks down when the dominant signal is *idiosyncratic*.

Example 15.1 (Earnings surprise)**.** Stock C reports earnings after the close. It is unobserved ($C \notin \mathcal{V}_{\text{obs}}$) because options have not yet traded. Meanwhile, index I and peer stock A are trading, reflecting the broad market.

The graph forces C to follow I and A: $\hat{\beta}_C \approx 0.36\,\hat{\beta}_I + 0.27\,\hat{\beta}_A + \cdots$. The earnings event—which should cause C's smile to steepen dramatically (uncertainty about the announcement)—is invisible to the system. Worse, if the market moves down overnight (dragging I and A with it), the graph will propagate a directional skew change to C that is *orthogonal* to the earnings effect.

More generally, the problem arises whenever the *prior is more informative than the propagated signal*: the node has a specific reason to differ from its neighbours, and the graph smoothing erases that specificity.

15.2 Dynamic self-trust

The solution is to make α_v event-aware. Define an *event schedule* $\mathcal{E} = \{(v, t, \delta\alpha)\}$ specifying, for each node v, event times t and self-trust increments $\delta\alpha$.

For a node with a pending idiosyncratic event:

$$\alpha_v^{\text{eff}} = \min\bigl(1,\ \alpha_v^{\text{base}} + \delta\alpha(v,t)\bigr), \tag{47}$$

where $\delta\alpha$ depends on the event type and its temporal proximity.

Event type	$\delta\alpha$	**Decay**
Earnings announcement	+0.40 to +0.60	Step-down post-release
Dividend ex-date	+0.20 to +0.30	Step-down post-ex
M&A rumour / pending	+0.30 to +0.50	Gradual over 2–5 days
Index rebalance	+0.15 to +0.25	Step-down post-rebalance

When α_v^{eff} is high, the row sum $\sum_w W_{vw}$ is low, and the node barely moves from its prior regardless of what the graph neighbours do. Once the event resolves (earnings are released, the stock starts trading), the self-trust drops back and the node re-enters the normal propagation regime.

Trader's Intuition

This is the mathematical version of what every vol trader does intuitively: "AAPL reports tonight, so I'm not going to blindly move AAPL vol just because SPX moved—I'm going to wait and see." The dynamic α_v formalises this instinct and applies it systematically across the entire universe.

15.3 Event-conditional priors

For earnings events specifically, the prior itself should reflect the expected vol jump. If the at-the-money implied variance for stock C decomposes as:

$$\sigma_C^2 T \;=\; \sigma_{\text{ex-earn}}^2 T \;+\; \sigma_{\text{jump}}^2, \tag{48}$$

then the prior $\psi_C^{(0)}$ should be calibrated to the *ex-earnings* surface, not the full surface including the earnings premium. The jump component σ_{jump}^2 is stripped out before the propagation and added back afterwards. This prevents the graph from mistakenly "propagating away" the earnings premium by smoothing it toward the peers.

16 Calendar Arbitrage: The Repaired-Mark Misfit

16.1 The problem

The post-hoc convex-order projection (Section 10.5) operates outside the optimisation. If the graph propagation creates a significant calendar violation—common in equity markets with steep term structures or lumpiness from discrete dividends—the projection will flatten the term structure.

The conflict: after projection, the surface no longer matches the market quotes y_v that drove the initial solve. The data fidelity has been sacrificed to satisfy an arbitrage constraint that was invisible to the optimiser. For large violations, the resulting implied volatility surface can look visibly different from the market, undermining trader confidence in the marks.

Implementation Warning

This is not an academic concern. In equity vol, calendar violations of 0.3–0.5 vol points are routine around ex-dividend dates, and violations of 1–2 points can occur when short-dated vol collapses (e.g. post-FOMC expiry crushing). A post-hoc fix of this magnitude is clearly visible on the trading screen.

16.2 Solution: Calendar constraints in the QP

The calendar condition (41), discretised on a grid $\{u_j\}_{j=1}^{J}$, gives a set of linear inequalities in the basis coefficients:

$$\sum_{j=1}^{k} \big(Q_{i,T_{n+1}}(u_j) - Q_{i,T_n}(u_j)\big)\,\Delta u \;\geq\; 0 \qquad \forall\, k = 1, \ldots, J, \tag{49}$$

for every asset i and consecutive maturity pair (T_n, T_{n+1}).

Since $Q_{i,T}(u) = m_{i,T} + s_{i,T}\widetilde{Q}_{i,T}(u)$ and the shape perturbation is linear in $\boldsymbol{\beta}$, these constraints are (to first order) linear in the decision variables. Stacking them across all assets and maturities gives a sparse inequality system:

$$\mathbf{G}\,\boldsymbol{\beta} \;\geq\; \mathbf{h}, \tag{50}$$

where $\mathbf{G}$ encodes the cumulative quantile differences and $\mathbf{h}$ absorbs the prior's existing convex-order margin.

The full constrained QP is then:

$$\boxed{\min_{\boldsymbol{\beta}}\; J(\boldsymbol{\beta}) \quad \text{s.t.} \quad \begin{cases} \beta_{v,1} \geq \epsilon, \;\; \beta_{v,2} \geq \epsilon & \text{(tail integrability)} \\ \mathbf{c}_v^\top \beta_v = 0 & \text{(unit variance, Section 13)} \\ \mathbf{d}_v^\top \beta_v = 0 & \text{(martingale, Section 13)} \\ \mathbf{G}\,\boldsymbol{\beta} \geq \mathbf{h} & \text{(calendar monotonicity)} \end{cases}} \tag{51}$$

Practical Remark

The number of constraints in (51):

- Tail integrability: $2|\mathcal{V}|$ box constraints (trivial).
- Unit variance + martingale: $2|\mathcal{V}|$ equality constraints.
- Calendar: $N_{\text{assets}} \times (N_T - 1) \times J$ inequality constraints. For 200 assets, 10 maturities, $J = 50$ grid points: $200 \times 9 \times 50 = 90{,}000$ inequalities. This is large but very sparse (each constraint involves only the M coefficients of two adjacent maturity nodes).

A sparse QP solver like OSQP handles this problem size in ∼20–50ms with warm-starting. The key is that most calendar constraints are inactive (the prior already satisfies them); only the violated or near-violated ones affect the solution.

16.3 Graceful degradation

For systems where the full QP is too expensive (e.g. ultra-low-latency requirements), a two-tier approach works well:

1. **Normal regime:** Solve the unconstrained system (29) with post-hoc PAVA projection. Calendar violations are rare and small; the projection is cosmetic.
2. **Stress regime:** When the post-hoc projection exceeds a threshold (e.g. > 0.3 vol points at any node), switch to the full QP (51) for that asset's maturity ladder only. This limits the QP to $\sim N_T \times M \sim 50$ variables with $\sim N_T \times J \sim 500$ constraints—trivially fast.

17 Extensions and Open Questions

17.1 Separate graphs for scale and shape

The framework applies the same graph $\mathbf{W}$ to all basis coefficients. A natural extension uses different weights for the scale component s vs. the shape $\widetilde{Q}$. Scale propagation is driven by market-wide factors; shape propagation is more idiosyncratic.

17.2 Time-varying graphs

During stress events, correlations spike and the graph should tighten (more influence from the index, less idiosyncrasy). A regime-dependent $\mathbf{W}(\text{VIX level})$ would adapt the propagation strength to market conditions.

17.3 Anti-correlation and negative weights

Some asset pairs are negatively correlated (e.g. equity vs. rates). The current framework requires $W_{vw} \geq 0$. To handle anti-correlation, one can use $|W_{vw}|$ in the graph and flip the sign of the propagated shock, or introduce a separate signed-influence term outside the $\mathbf{L}^\top\mathbf{L}$ structure.

17.4 Bayesian uncertainty quantification

The Tikhonov term $\eta\|\boldsymbol{\beta}\|^2$ is equivalent to a Gaussian prior on $\boldsymbol{\beta}$. A full Bayesian treatment provides posterior uncertainty bands on the reconstructed surfaces: "how uncertain is our mark for this illiquid stock?" The posterior covariance is the inverse of the Hessian in (29)—already available as a byproduct of the solve.

17.5 Market-implied graph calibration

The weights in $\mathbf{W}$ can be calibrated from data:

- **Historical**: rolling correlations, PCA loadings, or Granger causality of IV time series.
- **Market-implied**: infer $\mathbf{W}$ by minimising out-of-sample prediction error (leave-one-out cross-validation on surfaces).
- **Structural**: index composition, sector membership, known economic relationships.

A hybrid combining all three is likely optimal.

18 Implementation Roadmap

For teams adopting this framework, we recommend a phased rollout:

Phase 1: Prototype (2–4 weeks). Single maturity, 20–50 assets, unconstrained CG solve with post-hoc PAVA. Goal: validate the propagation matrix and calibrate λ by leave-one-out backtesting.

Phase 2: Full term structure (4–8 weeks). All maturities, dynamic self-trust from event schedule, tail-specific regularisation. Calendar constraints handled post-hoc. Goal: replace the existing marking engine for a single desk.

Phase 3: Constrained QP (4–6 weeks). Integrate OSQP or a comparable sparse QP solver. Add the linearised martingale, unit-variance, and calendar constraints. Goal: robustness under stress scenarios without post-hoc patching.

Phase 4: Bayesian extension + graph calibration (ongoing). Posterior covariance for mark uncertainty. Market-implied graph weights via cross-validation. Goal: uncertainty-aware marks and adaptive propagation.

Practical Remark

The most impactful early win is Phase 1: a working propagation matrix that visibly improves the consistency of marks across correlated names. This can be demonstrated to the trading desk in weeks, not months, and builds the credibility needed to justify the more complex phases.

Conclusion

This framework provides a principled, scalable approach to synchronous volatility marking by:

1. **Propagating full distributions**, not pointwise vols, using the Wasserstein geometry of quantile functions.
2. **Guaranteeing butterfly no-arbitrage** by construction through the LQD parametrisation.
3. **Encoding cross-asset and term-structure dependencies** in a directed graph whose harmonic extension provides a closed-form propagation rule.
4. **Solving a single convex optimisation** that balances data fidelity, graph consistency, and smoothness, with optional constraints for calendar arbitrage, martingale compliance, and tail integrability.

The four structural tensions identified in Part II—the re-normalisation feedback loop, the basis boundary paradox, the structural causality gap, and the calendar repair misfit—are not weaknesses of the framework but inherent features of the financial engineering problem. Each admits a concrete solution: manifold-constrained QP for the feedback loop, tail-specific regularisation for the boundary paradox, dynamic self-trust for the causality gap, and integrated calendar constraints for the misfit.

The result is a system that is both theoretically sound—rooted in optimal transport geometry and convex optimisation—and practically deployable within the latency and robustness requirements of a real-time derivatives trading desk.